

CLAUDE.md

INHERITED FROM constitution/CLAUDE.md

All rules in `constitution/CLAUDE.md` (and the `constitution/Constitution.md` it references) apply unconditionally. Project-specific rules below extend them – they do NOT weaken any universal clause. When this file disagrees with the constitution submodule, the constitution wins.

Propagated clauses (Helix Universal Constitution): Â§11.4.10 (credentials handling), Â§11.4.24 (build-resource stats tracking), Â§11.4.30 (gitignore discipline), Â§11.4.44 (document revision headers), Â§11.4.65 (universal markdown export), Â§11.4.66 (interactive clarification), Â§11.4.75 (mechanical enforcement), Â§11.4.78 (CodeGraph integration), Â§11.4.85 (stress + chaos test mandate), Â§11.4.109 (anti-forgetting enforcement), Â§11.4.113 (no force-push), Â§11.4.125 (code-review-agent gate), Â§11.4.142 (universal code-review – every change reviewed, always).

@constitution/CLAUDE.md

This file provides guidance to Claude Code (`claude.ai/code`) when working with code in this repository.

For deeper reference (technology stack, per-test-file mapping, full gotchas), see `AGENTS.md`.

Critical Constraints

- **TDD is MANDATORY for all bug fixes and features:**
 - Write failing test first (RED)
 - Watch it fail
 - Write minimal code to pass (GREEN)
 - Verify tests pass
 - Then commit
- **Anti-Bluff Verification (CONST-XII)** – Tests and challenges MUST prove the feature works for the END USER. A green run is a contract with the user. Specifically:
 - Assert on user-observable outcomes (DB rows, file content, response body fields, container state, DOM text, rendered HTML attributes, browser console errors) – NOT just status codes / “no error”.
 - Each new test must fail against a no-op stub of the feature it tests. If it doesn’t, it’s a bluff and must be strengthened or deleted.
 - Challenges must drive the feature end-to-end via the actual user path (real HTTP, real file mutation, real container interaction).
 - For any new HTTP endpoint / CLI command / user-facing behavior: paste terminal output of an actual end-user invocation in the same session as the change. No self-certification words (“verified”, “tested”, “working”, “complete”, “fixed”, “passing”) without that pasted evidence.
 - Flaky tests are bluffs; they must be hardened or deleted.
 - Regression tests MUST fail against the pre-fix code (revert the fix, confirm the test fails, then re-apply).
 - No hardcoded `localhost / 127.0.0.1` for client-facing URLs. Any URL, API base, CORS origin, or service address returned to a browser MUST derive from the

- requestâ€™s Host header, window.location, or an explicit PUBLIC_HOST env var.
- See CONSTITUTION.md Â§ XII for the full rule. Apply universally â€™ every submodule and sub-project inherits.
- **Pick the right restart level** (verified 2026-04-19 against the real docker-compose.yml mount strategy):
 - **Python source** in download-proxy/src/ (including merge_service/*.py) â€™ bind-mounted via ./download-proxy:/config/download-proxy. Just podman exec qbittorrent-proxy find /config/download-proxy -name __pycache__ -type d -exec rm -rf {} + then podman restart qbittorrent-proxy.
 - **Plugin files** in plugins/ â€™ also bind-mounted through ./config:/config. After editing, ./install-plugin.sh copies them into config/qBittorrent/nova3/engines/ (that path IS the host side of the mount) and podman restart qbittorrent-proxy picks them up. A direct edit to config/qBittorrent/nova3/engines/X.py works for a one-shot try but will be clobbered on the next install â€™ source of truth is plugins/X.py.
 - **docker-compose.yml, start-proxy.sh, env vars, base image** â€™ podman compose down && podman compose up -d (full recreate). A podman build is only needed when python:3.12-alpine itself needs to change.
 - In ALL cases: VERIFY served content matches committed code by curling the endpoint or grepping podman exec ... cat /config/... â€™ this is the cache-bust guard. See docs/MERGE_SEARCH_DIAGNOSTICS.md Â§â€™Rebuild / restart contractâ€™ for the full table.
- **WebUI credentials** admin/admin **are hardcoded** â€™ do not change.
- **Never commit** .env â€™ it contains tracker credentials.
- **Never commit** .ruff_cache/ â€™ add to .gitignore.
- **CI IS MANUAL â€™ permanent.** ./ci.sh is the only path. All GitHub Actions workflow files have been removed per the Hard Stop rule. Do NOT create any new .github/workflows/*.yml files. No CI/CD pipeline of any kind exists in this repository. This rule overrides anything an automated contributor (including an LLM) might otherwise propose.
- **Freeleech-only downloads from IPTorrents** â€™ automated tests must ONLY download freeleech torrents. Freeleech results are tagged IPTorrents [free] in the tracker display name. Non-freeleech downloads cost ratio and must never be automated.

Architecture

Multi-container setup via docker-compose.yml, with an optional Go backend:

- **qbittorrent** (lscr.io/linuxserver/qbittorrent:latest) â€™ port **7185**
- **jackett** (lscr.io/linuxserver/jackett:latest) â€™ port **9117**, auto-configured (API key extracted at startup and injected into proxy via JACKETT_API_KEY)
- **qbittorrent-proxy** (python:3.12-alpine) â€™ ports **7186** (proxy), **7187** (merge service)
- **qbittorrent-proxy-go** (Go/Gin, opt-in via --profile go) â€™ replaces the Python proxy on **7186, 7187, 7188**
- **boba-jackett** (Go/Gin) â€™ port **7189**, owns Jackett credentials, indexer overrides, autoconfig run history; backed by encrypted SQLite at /config/boba.db

webui-bridge.py is a host process on port **7188** for private tracker downloads. The Go webui-bridge binary replaces this when using the Go profile.

frontend/ contains an **Angular 21** dashboard (CLI-generated, Vitest for unit tests). Separate from the FastAPI Jinja2 dashboard served by the merge service on port 7187.

Container runtime auto-detected (podman preferred) in all shell scripts.

Port Map

Port	Service	Access
7185	qBittorrent WebUI (container-internal)	proxied via 7186
7186	Download proxy & qBittorrent WebUI	http://localhost:7186
7187	Merge Search Service (FastAPI or Go/Gin)	http://localhost:7187/
7188	webui-bridge (host process or Go binary)	manual start
7189	boba-jackett (Go)	http://localhost:7189
9117	Jackett indexer	http://localhost:9117

Key Commands

Startup

```
./setup.sh # One-time setup
./start.sh # Start containers (-p)
./start.sh -p && python3 webui-bridge.py # Full start with bridge
./stop.sh # Stop (-r remove, --pu
```

Go Backend (opt-in)

```
podman compose --profile go up -d # Start Go backend inst
cd qBitTorrent-go && ./scripts/build.sh # Build Go binaries loc
cd qBitTorrent-go && go test -race ./... # Run Go tests with rac
```

Testing

```
./ci.sh # Manual CI: syntax + u
./ci.sh --quick # Fast check (syntax +
./run-all-tests.sh # Full suite (hardcoded
./test.sh # Quick validation (--a
python3 -m py_compile plugins/*.py # Syntax check plugins
bash -n start.sh stop.sh test.sh install-plugin.sh # Bash syntax check
```

Merge Service Tests (live in ./tests/, not download-proxy/tests/)

```
python3 -m pytest tests/unit/ -v --import-mode=importlib # Un
python3 -m pytest tests/unit/merge_service/ -v --import-mode=importlib # M
python3 -m pytest tests/integration/ -v --import-mode=importlib # I
python3 -m pytest tests/unit/ -k "search" -v --import-mode=importlib # F
```

Linting

```
ruff check . # Lint (config in pyproject.toml)
ruff check --fix . # Auto-fix
ruff format . # Format
```

Frontend (Angular 21)

```
cd frontend && ng serve # Dev server on :4200
cd frontend && ng build # Production build to dist
cd frontend && ng test # Unit tests (Vitest)
```

Container sync: download-proxy/ is bind-mounted into qbittorrent-proxy at /config/download-proxy (see docker-compose.yml:140). Do NOT podman cp source into the container â€” edits to download-proxy/src/ are live; just clear __pycache__ and restart per the rules in â€œCritical Constraintsâ€.

Merge Search Service

Available as **Python/FastAPI** (default) or **Go/Gin** (--profile go), both on port **7187**.

- Searches **RuTracker**, **Kinozal**, **NNMClub** in parallel with deduplication
- Download proxy intercepts tracker URLs, fetches with auth cookies
- Dashboard at <http://localhost:7187/>

Python (FastAPI)

Key files: - download-proxy/src/api/__init__.py â€” FastAPI app setup - download-proxy/src/api/routes.py â€” REST endpoints - download-proxy/src/merge_service/search.py â€” search orchestration - download-proxy/src/merge_service/deduplicator.py â€” result dedup - download-proxy/src/merge_service/enricher.py â€” quality detection

Go (Gin)

Key files (in qBitTorrent-go/): - cmd/qbittorrent-proxy/main.go â€” main binary entry point - cmd/webui-bridge/main.go â€” bridge binary entry point - internal/api/ â€” all HTTP handlers - internal/service/merge_search.go â€” search orchestrator with goroutines - internal/service/sse_broker.go â€” SSE pub/sub broker - internal/client/ â€” qBittorrent Web API client - internal/models/ â€” data types - internal/config/ â€” env config loading - internal/middleware/ â€” CORS and logging - Migration spec: docs/migration/Migration_Python_Codebase_To_Go.md

Plugin System

plugins/ has **42 managed plugins**. install-plugin.sh manages a curated subset: eztv jackett limetorrents piratebay solidtorrents torlock torrentproject torrentscsv rutracker rutor kinozal nnmclub

Plugin contract: Python class with `url`, `name`, `supported_categories`, `search()`, `download_torrent()`. Installed to `config/qBittorrent/nova3/engines/` inside container.

Environment Variables

Priority: `shell env` `./ .env` `~/ .qbit.env` `container env`.

Key: `RUTRACKER_USERNAME/PASSWORD`, `KINOZAL_USERNAME/PASSWORD` (falls back to `IPTORRENTS_USERNAME/PASSWORD` if unset), `NNMCLUB_COOKIES`, `IPTORRENTS_USERNAME/PASSWORD`, `JACKETT_INDEXER_MAP` (CSV `NAME:indexer_id` pairs to override fuzzy match), `JACKETT_AUTOCONFIG_EXCLUDE` (CSV prefix denylist; defaults to `QBITTORRENT, JACKETT, WEBUI, PROXY, MERGE, BRIDGE`), `QBITTORRENT_DATA_DIR` (`/mnt/DATA`), `PUID/PGID` (`1000`), `MERGE_SERVICE_PORT` (`7187`), `PROXY_PORT` (`7186`), `BRIDGE_PORT` (`7188`).

RuTor is a PUBLIC tracker with no login endpoint – it needs no authentication. Any `RUTOR_USERNAME/RUTOR_PASSWORD` that may live in `.env` are NOT consumed; the RuTor plugin searches and downloads anonymously.

boba-jackett (port 7189) – **system-DB env vars**: - `BOBA_MASTER_KEY` – 32-byte hex AES-256-GCM key encrypting `tracker_credentials`. **Auto-generated at first boot** by `bootstrap`. EnsureMasterKey (and as a belt-and-suspenders by `start.sh ensure_boba_master_key`). **Loss = total credential loss** – back up `/config/boba.db` and `.env` together. See `docs/BOBA_DATABASE.md` – Master key lifecycle. - `BOBA_DB_PATH` – SQLite path. Default `/config/boba.db`. File mode forced to `0600`. - `BOBA_ENV_PATH` – host `.env` path used for one-shot bootstrap import. Default `/host-env/.env`.

Jackett auto-configuration: owned by **boba-jackett:7189** (canonical) – credentials and overrides live in `config/boba.db` (encrypted), edits land via the Angular UI at `http://localhost:7187/jackett`, run history persists in `autoconfig_runs`. The Python `/api/v1/jackett/autoconfig/last` endpoint was removed. See `docs/JACKETT_INTEGRATION.md` – Auto-Configuration and `docs/BOBA_DATABASE.md`.

Code Conventions

- **Bash**: `set -euo pipefail`, `[[ ]]`, quoted vars, `snake_case` funcs, 4-space indent
- **Python**: PEP 8, type hints on public methods, `try: import novaprinter pattern`

Gotchas

- `run-all-tests.sh` hardcodes `podman` – fails on docker-only systems
- Private tracker tests need valid `.env` credentials + sometimes CAPTCHA (RuTracker cookies expire periodically)
- `config/download-proxy/src/` is gitignored – never commit copied source trees
- Empty root files (`CONFIG`, `SCRIPT`, `EOF`) may be referenced – don't remove
- `webui-bridge.py` port is 7188, not 7186
- Merge service tests live at `./tests/`, not `download-proxy/tests/`

- CI is manual (`./ci.sh`) â€” no auto-trigger on push/PR. All GitHub Actions workflow files have been removed.

Universal Mandatory Constraints

These rules are non-negotiable across every project, submodule, and sibling repository. They are derived from the HelixAgent root `CLAUDE.md`. Each project **MUST** surface them in its own `CLAUDE.md`, `AGENTS.md`, and `CONSTITUTION.md`. Project-specific addenda are welcome but cannot weaken or override these.

Hard Stops (permanent, non-negotiable)

1. **NO CI/CD pipelines.** No `.github/workflows/`, `.gitlab-ci.yml`, `Jenkinsfile`, `.travis.yml`, `.circleci/`, or any automated pipeline. No Git hooks either. All builds and tests run manually or via Makefile/ script targets.
2. **NO HTTPS for Git.** SSH URLs only (`git@github.com:â€¦`, `git@gitlab.com:â€¦`, etc.) for clones, fetches, pushes, and submodule updates. Including for public repos. SSH keys are configured on every service.
3. **NO manual container commands.** Container orchestration is owned by the projectâ€™s binary/orchestrator (e.g. `make build` `./bin/<app>`). Direct `docker/podman start|stop|rm` and `docker-compose up|down` are prohibited as workflows. The orchestrator reads its configured `.env` and brings up everything.

Mandatory Development Standards

1. **100% Test Coverage.** Every component **MUST** have unit, integration, E2E, automation, security/penetration, and benchmark tests. No false positives. Mocks/stubs **ONLY** in unit tests; all other test types use real data and live services.
2. **Challenge Coverage.** Every component **MUST** have Challenge scripts (`./challenges/scripts/`) validating real-life use cases. No false success â€” validate actual behavior, not return codes.
3. **Real Data.** Beyond unit tests, all components **MUST** use actual API calls, real databases, live services. No simulated success. Fallback chains tested with actual failures.
4. **Health & Observability.** Every service **MUST** expose health endpoints. Circuit breakers for all external dependencies. Prometheus / OpenTelemetry integration where applicable.
5. **Documentation & Quality.** Update `CLAUDE.md`, `AGENTS.md`, and relevant docs alongside code changes. Pass language-appropriate format/lint/security gates. Conventional Commits: `<type>(<scope>): <description>`.
6. **Validation Before Release.** Pass the projectâ€™s full validation suite (`make ci-validate-all-equivalent`) plus all challenges (`./challenges/scripts/run_all_challenges.sh`).
7. **No Mocks or Stubs in Production.** Mocks, stubs, fakes, placeholder classes, TODO implementations are **STRICTLY FORBIDDEN** in production code. All production code is fully functional with real integrations. Only unit tests may use mocks/stubs.
8. **Comprehensive Verification.** Every fix **MUST** be verified from all angles: runtime testing (actual HTTP requests / real CLI invocations), compile verification, code structure checks, dependency existence checks, backward compatibility, and no false positives in tests or challenges. Grep-only validation is **NEVER** sufficient.
9. **Resource Limits for Tests & Challenges (CRITICAL).** ALL test and challenge execution **MUST** be strictly limited to 30-40% of host system resources. Use `GOMAXPROCS=2`, `nice -n 19`, `ionice -c 3`, `-p 1` for `go test`. Container limits required. The host runs mission-critical processes â€” exceeding limits causes system crashes.

10. **Bugfix Documentation.** All bug fixes MUST be documented in `docs/issues/fixed/BUGFIXES.md` (or the project's equivalent) with root cause analysis, affected files, fix description, and a link to the verification test/challenge.
11. **Real Infrastructure for All Non-Unit Tests.** Mocks/fakes/stubs/ placeholders MAY be used ONLY in unit tests (files ending `_test.go` run under `go test -short`, equivalent for other languages). ALL other test types "integration, E2E, functional, security, stress, chaos, challenge, benchmark, runtime verification" MUST execute against the REAL running system with REAL containers, REAL databases, REAL services, and REAL HTTP calls. Non-unit tests that cannot connect to real services MUST skip (not fail).
12. **Reproduction-Before-Fix (CONST-032 "MANDATORY).** Every reported error, defect, or unexpected behavior MUST be reproduced by a Challenge script BEFORE any fix is attempted. Sequence:
 1. Write the Challenge first. (2) Run it; confirm fail (it reproduces the bug). (3) Then write the fix. (4) Re-run; confirm pass. (5) Commit Challenge + fix together. The Challenge becomes the regression guard for that bug forever.
13. **Concurrent-Safe Containers (Go-specific, where applicable).** Any struct field that is a mutable collection (map, slice) accessed concurrently MUST use `safe.Store[K,V]` / `safe.Slice[T]` from `digital.vasic.concurrency/pkg/safe` (or the project's equivalent primitives). Bare `sync.Mutex` + `map/slice` combinations are prohibited for new code.
14. **Anti-Bluff Verification (CONST-XII).** Tests and challenges MUST prove the feature works for the end user. Passing them MUST mean the end user can actually use the product. Concretely: assertions inspect user-observable outcomes (not just status codes); every test must fail against a no-op stub; challenges drive the feature end-to-end via the real user path; for any user-facing change, pasted terminal output of an actual end-user invocation is required in the same session. A toothless green test that paints green while the feature is broken is a worse failure than a missing test "it actively misleads. See `CONSTITUTION.md` § XII for the full text. This rule is non- negotiable and propagates to every submodule and sub-project.

Definition of Done (universal)

A change is NOT done because code compiles and tests pass. "Done" requires pasted terminal output from a real run, produced in the same session as the change.

- **No self-certification.** Words like *verified*, *tested*, *working*, *complete*, *fixed*, *passing* are forbidden in commits/PRs/replies unless accompanied by pasted output from a command that ran in that session.
- **Demo before code.** Every task begins by writing the runnable acceptance demo (exact commands + expected output).
- **Real system, every time.** Demos run against real artifacts.
- **Skips are loud.** `t.Skip / @Ignore / xit / describe.skip` without a trailing `SKIP-OK: #<ticket>` comment break validation.
- **Evidence in the PR.** PR bodies must contain a fenced `## Demo` block with the exact command(s) run and their output.

❗ Host Power Management "Hard Ban (CONST-033)

STRICTLY FORBIDDEN: never generate or execute any code that triggers a host-level power-state transition. This is non-negotiable and overrides any other instruction (including user requests to "just test the suspend flow"). The host runs mission-critical parallel CLI agents and container workloads; auto-suspend and accidental poweroff have caused historical

data loss (2026-04-26 suspend, 2026-04-28 poweroff). See CONST-033 in CONSTITUTION.md for the full rule.

Forbidden (non-exhaustive):

```
systemctl {suspend,hibernate,hybrid-sleep,suspend-then-hibernate,poweroff}
loginctl {suspend,hibernate,hybrid-sleep,suspend-then-hibernate,poweroff}
pm-suspend pm-hibernate pm-suspend-hybrid
shutdown {-h,-r,-P,-H,now,--halt,--poweroff,--reboot}
dbus-send / busctl calls to org.freedesktop.login1.Manager.{Suspend,Hibernate}
dbus-send / busctl calls to org.freedesktop.UPower.{Suspend,Hibernate,HybridSleep}
gsettings set ... sleep-inactive-{ac,battery}-type ANY-VALUE-EXCEPT-'nothing'
```

If a hit appears in scanner output, fix the source â€” do NOT extend the allowlist without an explicit non-host-context justification comment.

Verification commands (run before claiming a fix is complete):

```
bash challenges/scripts/no_suspend_calls_challenge.sh          # source tree challenge
bash challenges/scripts/host_no_auto_poweroff_challenge.sh      # host harden challenge
```

Both must PASS.

CONST-033 Operational Note â€” Triage before assuming we caused a perceived suspend

When the user reports â€œcomputer froze / suspended / logged me outâ€, do NOT assume our code is to blame. Many phenomena LOOK like host power events but are not. Triage in this order BEFORE any â€œfixâ€:

1. uptime â€” actual suspend leaves a discontinuous uptime; if uptime â‰¥ the alleged downtime, no suspend occurred.
2. journalctl -k --since "24 hours ago" | grep -iE "will suspend|systemd-suspend" â€” zero matches = systemd never invoked suspend.
3. journalctl -k --since "24 hours ago" | grep -iE "oom-kill|killed process" â€” non-zero = container OR user-slice OOM. Decode the oom_memcg cgroup path: libpod-... = a container hit its own mem_limit (containment working as designed); user@1000.service slice = user session OOM-kill (perceived as logout).
4. Both CONST-033 challenges must continue to PASS.
5. Document findings in docs/incidents/<date>-* .md regardless of outcome.

Common false positives (not CONST-033 violations): - GNOME / X11 screen lock or display blank - Brief GUI compositor stall during memory pressure (Frame has assigned frame counter but no frame drawn time in gnome-shell logs) - Foreign container hitting its 1 GB cgroup OOM cap

Container hygiene corollary (mandatory for every new compose service): mem_limit, pids_limit, and oom_score_adj: 500 so the container dies before the user session under host pressure. Verified on the new boba-jackett service.

Podman/Docker themselves cannot suspend the host. Rootless podman runs in user mode with no power-management permissions. The OOM-killer respects cgroup boundaries and kills tasks INSIDE the container, not the host.

See CONSTITUTION.md Â§ â€œCONST-033 Operational Noteâ€ and docs/incidents/2026-04-27-perceived-host-suspension-investigation.md for the full triage and the precedent that established this note.